

Methods: Sombrero

ALMA Thesis Planner (automated pipeline)

1. Archive inventory and epoch audit (weeks 1–2)

Begin at the ALMA Science Archive with project code 2011.0.00754.S. Download the full delivery — raw ASDMs, calibration scripts, QA2 products, and any archive-delivered calibrated measurement sets. Before touching any data, build a spreadsheet with one row per scheduling block: the frequency setting (spectral window central frequencies), the band, the execution date and time (from the ASDM metadata or the archive’s execution-block listing), on-source integration time, number of antennas, and the QA2 status. This is your executed-versus-proposed census: the proposal requested 15 frequency settings across Bands 3, 6, 7, and 9, and Cycle 0 programs were routinely only partially observed, so establish exactly which settings exist and are calibratable. Do not assume any setting exists until you see its execution block.

From the same table, compute the epoch spread. Group settings executed within a few days of one another into a single epoch. Any setting executed weeks or months from the main epoch gets flagged now: for those frequencies you will later inflate the flux-density uncertainty by a variability term rather than treat the point as simultaneous with the rest. Estimate that term from the nuclear light curve within the dataset itself where possible — if any frequency setting was observed at two epochs, the flux difference between them is your empirical variability amplitude; otherwise adopt a conservative fractional variability floor and state it explicitly in the error budget. The deliverable of this step is a table you will reproduce verbatim in the thesis: frequency, band, date, integration time, epoch group, and calibration pathway.

Resolve the software pathway in the same two weeks. The delivered reduction scripts target CASA 3.x/4.x. Test three options in order and commit to the first that works: (a) archive-delivered calibrated measurement sets, verified by plotting calibrated visibility amplitudes of the flux and phase calibrators and checking them against the values recorded in the QA2 report; (b) a pinned legacy CASA installation (4.x era) run inside a container or dedicated environment, executing the delivered scripts unmodified; (c) only if both fail, a port of the calibration steps to modern CASA, done one script at a time with amplitude comparisons against the QA2 report at each stage. Record the choice and its verification in your notebook; the whole downstream reduction depends on it being reproducible.

2. Calibration error budget for slope and curvature (weeks 2–4)

Before reducing anything, write down the systematic error budget, because it defines what the spectrum can and cannot say. Adopt the Cycle 0 absolute flux-scale accuracies: ~5% in Band 3, ~10% in Bands 6 and 7, and 10–20% in Band 9. Then propagate. For a power law $S_\nu \propto \nu^\alpha$ fit between two frequencies ν_1 and ν_2 with fractional flux-scale errors f_1 and f_2 , the flux-scale contribution to the index error is $\sigma(\alpha) = \sqrt{f_1^2 + f_2^2} / \ln(\nu_2/\nu_1)$. Over the Band 3-to-Band 7 lever arm (~100–350 GHz, $\ln(\nu_2/\nu_1) \approx 1.25$) with 5% and 10% scales, this gives $\sigma(\alpha) \approx 0.09$ — call it ± 0.1 . Do this calculation yourself in a short Python script, then extend it: simulate